CSC 370 PaperTrade, Relational Algebra Constraints

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Overview

This document gives the relational algebra form of the `stock_trading` constraints. Constraint IDs match `docs/Schemata Overview/schemata.md` and the comments in `sql/ddl.sql`. Each constraint is written in one of two forms:

- **Subset** ($\subseteq$): referential integrity, where every value in one relation must appear in another.
- **Empty set** ($= \emptyset$): domain rules, where selecting the violating tuples must return nothing.

Constraints with no meaningful relational algebra form are left out. These are: primary keys and `UNIQUE` (covered in the Candidate Keys document); `NOT NULL`; `DEFAULT`s; `ON DELETE` actions; and rules about how the database may change, which the triggers enforce rather than a property of a single state (C.7.6–C.7.9, C.8.6, C.8.8).

Glossary

ID	Relation	Constraint	Form
C.1.4	User	Email has a valid format	$= \emptyset$
C.1.5	User	Cash balance is non-negative	$= \emptyset$
C.2.2	Portfolio	Owner exists	$\subseteq$
C.2.5	Portfolio	Name is non-empty	$= \emptyset$
C.3.3	Exchange	Currency is a 3-letter code	$= \emptyset$
C.3.4	Exchange	Exchange code is uppercase	$= \emptyset$
C.4.2	Stock	Exchange exists	$\subseteq$
C.4.4	Stock	Ticker is uppercase	$= \emptyset$
C.5.2	StockPriceHistory	Stock exists	$\subseteq$
C.5.3	StockPriceHistory	Prices are positive	$= \emptyset$
C.5.4	StockPriceHistory	High is at least low	$= \emptyset$
C.5.5	StockPriceHistory	Open and close lie within low–high	$= \emptyset$
C.5.6	StockPriceHistory	Volume is non-negative	$= \emptyset$
C.5.7	StockPriceHistory	Price date is not in the future	$= \emptyset$
C.6.2	Watchlist	User exists	$\subseteq$
C.6.3	Watchlist	Stock exists	$\subseteq$
C.6.4	Watchlist	Target price, if set, is positive	$= \emptyset$
C.7.2	Transaction	Portfolio and stock exist	$\subseteq$
C.7.3	Transaction	Quantity is positive	$= \emptyset$
C.7.4	Transaction	Price is positive	$= \emptyset$
C.7.5	Transaction	Type is buy or sell	$= \emptyset$
C.7.10	Transaction	Execution time is not in the future	$= \emptyset$
C.8.2	LimitOrder	Portfolio and stock exist	$\subseteq$
C.8.3	LimitOrder	Target price and quantity are positive	$= \emptyset$
C.8.4	LimitOrder	Type is buy or sell	$= \emptyset$
C.8.7	LimitOrder	Every executed order has a matching transaction	$\subseteq$

1 user

- C.1.4 Email has a valid format
- C.1.5 Cash balance is non-negative

C.1.4 Email has a valid format

Contains `@` and a domain.

$$\sigma_{\text{email NOT LIKE '\%_@_\%._\%'}}(\text{User}) = \emptyset$$

C.1.5 Cash balance is non-negative

A user can't spend money they don't have.

$$\sigma_{\text{cash_balance}<0}(\text{User}) = \emptyset$$

2 portfolio

- C.2.2 Owner exists
- C.2.5 Name is non-empty

C.2.2 Owner exists

Every portfolio belongs to a registered user.

$$\pi_{\text{user_id}}(\text{Portfolio}) \subseteq \pi_{\text{user_id}}(\text{User})$$

C.2.5 Name is non-empty

$$\sigma_{\text{portfolio_name}=''}(\text{Portfolio}) = \emptyset$$

3 exchange

- C.3.3 Currency is a 3-letter code
- C.3.4 Exchange code is uppercase

C.3.3 Currency is a 3-letter code

ISO 4217, e.g. `USD`, `CAD`.

$$\sigma_{\text{CHAR_LENGTH(currency)}\neq 3}(\text{Exchange}) = \emptyset$$

C.3.4 Exchange code is uppercase

Compared case-sensitively.

$$\sigma_{\text{exchange_code}\neq\text{UPPER(exchange_code)}}(\text{Exchange}) = \emptyset$$

4 stock

- C.4.2 Exchange exists
- C.4.4 Ticker is uppercase

C.4.2 Exchange exists

Every stock is listed on a known exchange.

$$\pi_{\text{exchange_code}}(\text{Stock}) \subseteq \pi_{\text{exchange_code}}(\text{Exchange})$$

C.4.4 Ticker is uppercase

Compared case-sensitively.

$$\sigma_{\text{ticker}\neq\text{UPPER(ticker)}}(\text{Stock}) = \emptyset$$

5 stock_price_history

- C.5.2 Stock exists
- C.5.3 Prices are positive
- C.5.4 High is at least low
- C.5.5 Open and close lie within low–high
- C.5.6 Volume is non-negative
- C.5.7 Price date is not in the future

C.5.2 Stock exists

A price record can't exist without its stock (weak entity).

$$\pi_{\text{ticker}}(\text{StockPriceHistory}) \subseteq \pi_{\text{ticker}}(\text{Stock})$$

C.5.3 Prices are positive

$$\sigma_{\text{opening_price}\leq 0 \,\vee\, \text{closing_price}\leq 0 \,\vee\, \text{high_price}\leq 0 \,\vee\, \text{low_price}\leq 0}(\text{StockPriceHistory}) = \emptyset$$

C.5.4 High is at least low

$$\sigma_{\text{high_price}<\text{low_price}}(\text{StockPriceHistory}) = \emptyset$$

C.5.5 Open and close lie within low–high

$$\sigma_{\text{opening_price}<\text{low_price} \,\vee\, \text{opening_price}>\text{high_price} \,\vee\, \text{closing_price}<\text{low_price} \,\vee\, \text{closing_price}>\text{high_price}}(\text{StockPriceHistory}) = \emptyset$$

C.5.6 Volume is non-negative

$$\sigma_{\text{volume}<0}(\text{StockPriceHistory}) = \emptyset$$

C.5.7 Price date is not in the future

Evaluated against the current date; enforced by trigger.

$$\sigma_{\text{price_date}>\text{CURDATE()}}(\text{StockPriceHistory}) = \emptyset$$

6 `watchlist`

- C.6.2 User exists
- C.6.3 Stock exists
- C.6.4 Target price, if set, is positive

C.6.2 User exists

$$\pi_{\text{user_id}}(\text{Watchlist}) \subseteq \pi_{\text{user_id}}(\text{User})$$

C.6.3 Stock exists

$$\pi_{\text{ticker}}(\text{Watchlist}) \subseteq \pi_{\text{ticker}}(\text{Stock})$$

C.6.4 Target price, if set, is positive

`NULL` means no alert and is allowed.

$$\sigma_{\text{target_price}\neq\text{NULL} \,\wedge\, \text{target_price}\leq 0}(\text{Watchlist}) = \emptyset$$

7 `transaction`

- C.7.2 Portfolio and stock exist
- C.7.3 Quantity is positive
- C.7.4 Price is positive
- C.7.5 Type is buy or sell
- C.7.10 Execution time is not in the future

C.7.2 Portfolio and stock exist

$$\pi_{\text{portfolio_id}}(\text{Transaction}) \subseteq \pi_{\text{portfolio_id}}(\text{Portfolio})$$
$$\pi_{\text{ticker}}(\text{Transaction}) \subseteq \pi_{\text{ticker}}(\text{Stock})$$

C.7.3 Quantity is positive

$$\sigma_{\text{quantity}\leq 0}(\text{Transaction}) = \emptyset$$

C.7.4 Price is positive

$$\sigma_{\text{price}\leq 0}(\text{Transaction}) = \emptyset$$

C.7.5 Type is buy or sell

$$\sigma_{\text{transaction_type}\notin\{\text{'buy'},\text{'sell'}\}}(\text{Transaction}) = \emptyset$$

C.7.10 Execution time is not in the future

Evaluated against the current time; enforced by trigger.

$$\sigma_{\text{executed_at}>\text{NOW()}}(\text{Transaction}) = \emptyset$$

8 limit_order

- C.8.2 Portfolio and stock exist
- C.8.3 Target price and quantity are positive
- C.8.4 Type is buy or sell
- C.8.7 Every executed order has a matching transaction

C.8.2 Portfolio and stock exist

$$\pi_{\text{portfolio_id}}(\text{LimitOrder}) \subseteq \pi_{\text{portfolio_id}}(\text{Portfolio})$$
$$\pi_{\text{ticker}}(\text{LimitOrder}) \subseteq \pi_{\text{ticker}}(\text{Stock})$$

C.8.3 Target price and quantity are positive

$$\sigma_{\text{target_price} \leq 0 \,\vee\, \text{quantity} \leq 0}(\text{LimitOrder}) = \emptyset$$

C.8.4 Type is buy or sell

$$\sigma_{\text{order_type} \notin \{\text{'buy'},\text{'sell'}\}}(\text{LimitOrder}) = \emptyset$$

C.8.7 Every executed order has a matching transaction

The trade is recorded at the order's target price. Transaction attributes are renamed to match.

$$\pi_{\text{portfolio_id, ticker, order_type, quantity, target_price}}\big(\sigma_{\text{status='executed'}}(\text{LimitOrder})\big)$$
$$\subseteq \rho_{(\text{portfolio_id, ticker, order_type, quantity, target_price})}\big(\pi_{\text{portfolio_id, ticker, transaction_type, quantity, price}}(\text{Transaction})\big)$$